2024 at the Barbados Cloud Observatory: a missing winter, missing high clouds

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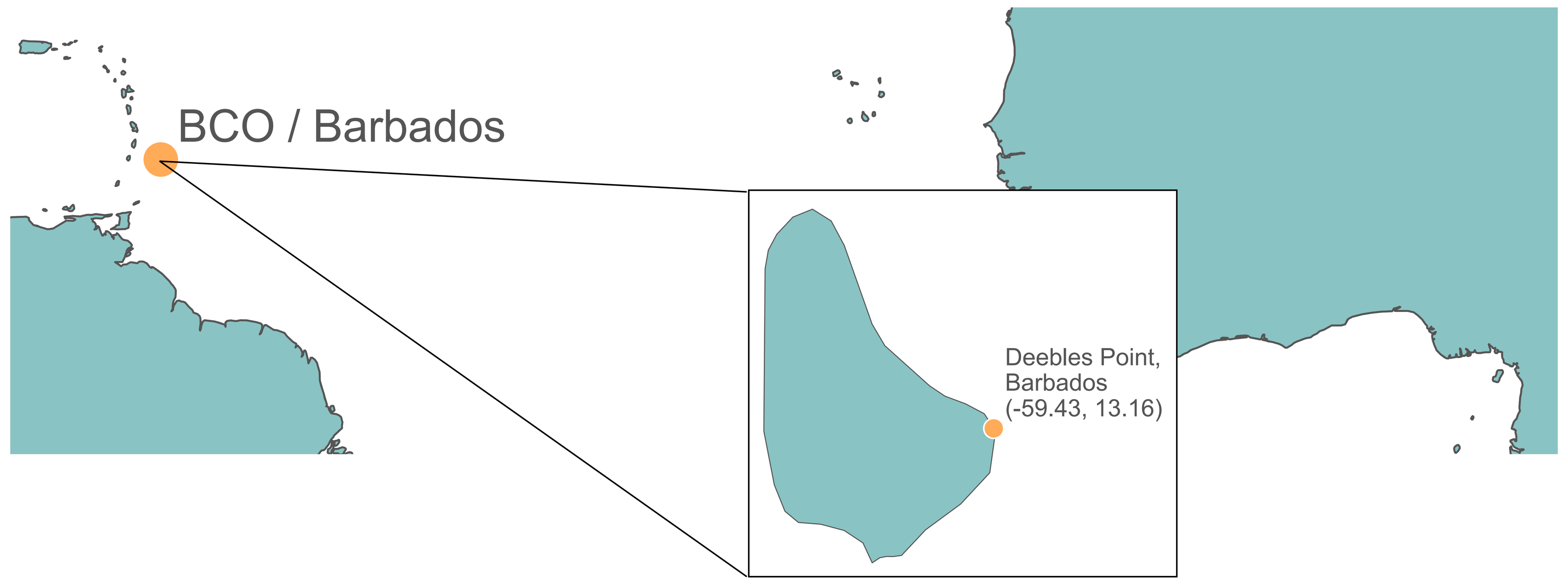

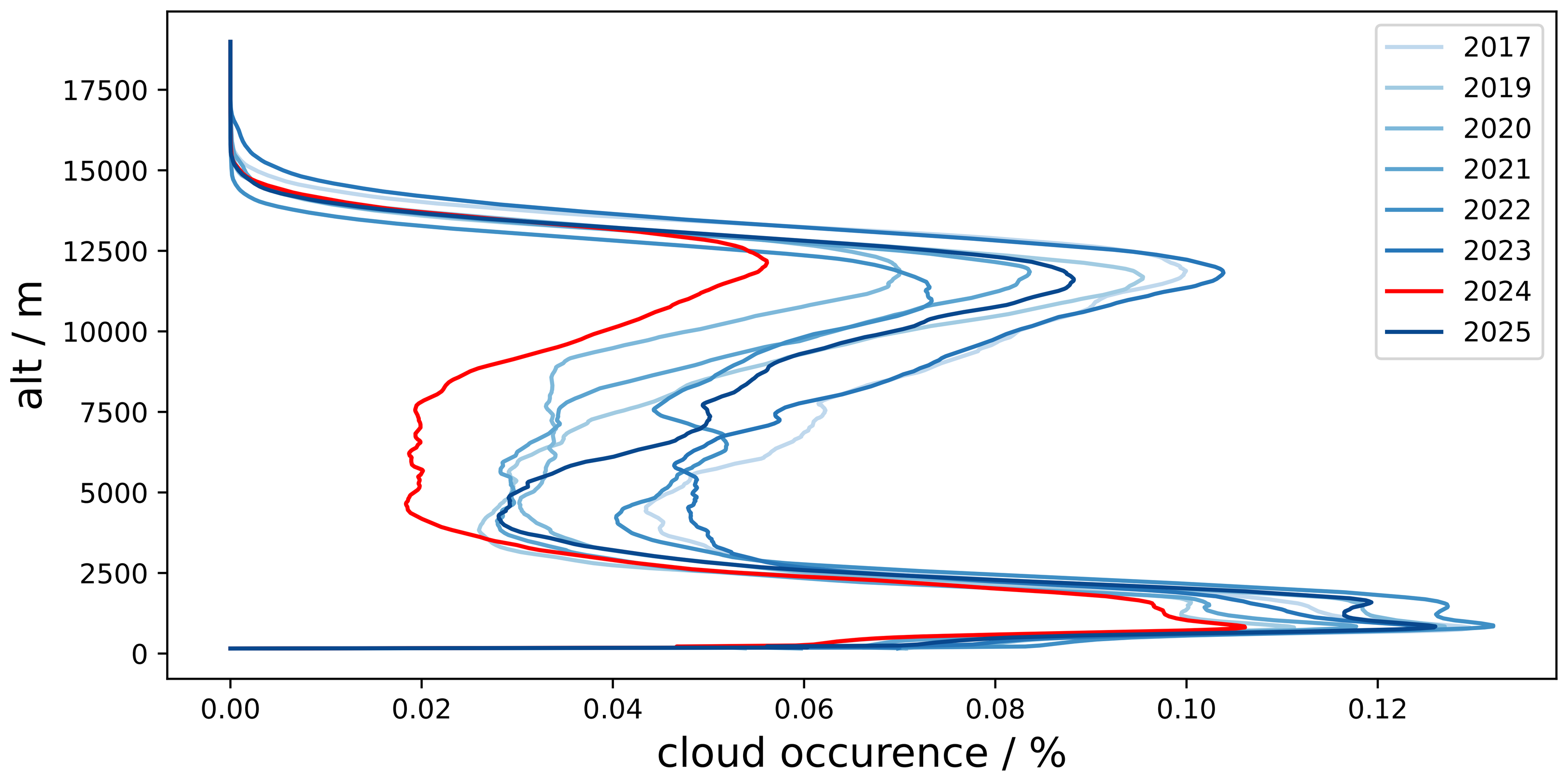

top: de-seasonalized, de-trended temperature time series:

BCO tas is a BCO 4-sensor near-surface temperature mean varying w.r.t. BCO mean

Bajan index is the ERA5 mean sst over longitudes in [-60, -40] and latitudes in [10, 20] varying w.r.t. the 1980-2010 climate

middle: monthly time series of a reflectivity mask (Ze > -40 dBZ), plotted at left as yearly averaged profiles, which show fewer clouds (particularly high clouds) in 2024

bottom: ratios of de-seasonalized, de-trended radiative fluxes to the standard deviation of their time series show increased downwelling sw, decreased sw diffuse, and increased lw diffuse in 2024

Open questions:

- which response drives which: clouds or SSTs?
- what teleconnections (if any) bridge the tropical Pacific El Niño and subtropical Atlantic SSTs?
- what is suppressing high cloud formation?
  - stability: LTS / EIS / z_{LCL}?
- what mechanisms govern the eventual cooling of SSTs and return of convection?

Scan for contact
rorlija1.github.io

Scan for data documentation
https://tcodata.mpimet.mpg.de/

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